# DIC17
# Latches and Flip-Flops

Specification-based laboratory report

17 October 2026

**English edition**
HSPICE S-2021.09 | Model library supplied separately

## 1 Objectives

Compare static, dynamic, and weak-feedback DFFs following handout PDF pages 74–77, covering hazards, noise immunity, frequency limits, and asynchronous reset. The circuits use transistors and TGs rather than ideal edge-triggered behavioral models.

## 2 Handout requirements and test mapping

| Requirement | Deck prefix and conditions | Status |
|---|---|---|
| Step 1: hazards, latch versus DFF | 01; 20 ns period | Waveforms measured |
| Step 1: operating-frequency limit | 03; 12 periods | Finite boundary located |
| Step 2: equal-noise comparison | 02; 1/5/20 fF | Measured |
| Fig. 17.1(c): weak feedback | 01/02 | Compared |
| Q1–3: overlap, slope, setup/hold | Theory; Lab21 sweep | Analyzed; not full extraction |
| Q4: reset-low DFF | 04; both phases | Measured |

## 3 Circuit and timing derivation

Two TG latches use complementary clocks: the master is transparent during low phase and the slave during high phase, producing a positive-edge capture path. The static circuit uses switched feedback at both storage nodes. The dynamic circuit removes feedback and stores charge parasitically. The weak-feedback circuit uses long-channel weak inverters. Feedback loading, restoration current, and contention with new data jointly affect speed.

Asynchronous reset replaces the slave output inverter with a NOR: $Q = \overline{Z + R}$, forcing low when $R = 1$. This verifies output reset, not a complete reset recovery/removal timing signoff.

# 4 Simulation setup and interpretation

The supplied encrypted MOS model is evaluated at a baseline supply of 1.0 V and 25 °C. The 1.0 V value is a teaching condition, not a verified voltage rating. These results do not reproduce the handout's original process or supply conditions. Unless stated in the deck, $W_n/W_p/L = 0.6/1.8/0.25\,\mu$m, bulks are tied to their corresponding rails, and output load is 5 fF. Transmission-gate and weak-feedback dimensions are specified explicitly in each deck.

PrimeSim HSPICE S-2021.09 win64 was used with `RUNLVL=6`, `ACCURATE`, and `WARNLIMIT=100`. Every baseline and `.alter` analysis concluded without errors or failed measurements. The simulator still reports suppressed warnings associated with encrypted blocks; this does not establish that the model is warning-free or that those warnings are harmless. The `.tran` print interval is not a fixed integration step: the solver uses adaptive timesteps. A finite sweep is not a complete process qualification.

Unless stated otherwise, the declared functional diagnostic uses $V \leq 0.2$ V for low and $V \geq 0.8$ V for high. These are report-level interpretation thresholds, not numerical handout requirements or extracted receiver noise margins. Correct samples do not exclude glitches between samples. Figures are redrawn from actual simulation data; `results/` contains per-case waveforms, measurements, and derived comparison CSV files.

Public decks omit model loading paths; model configuration is added only to local execution copies. The model itself is not distributed. The first two filename digits identify the deck numbers used below. Every deck is standalone.

# 5 Results

The three circuits produce approximately 1, 0, 0, and 0 V at 7, 27, 47, and 67 ns. Pulses during the closed phase do not directly propagate to the output. The 45 ns data transition deliberately lies near the sampling edge and the setup/hold boundary; one waveform cannot prove that all near-edge hazards are safely rejected.

| Coupling (fF) | Static peak (V) | Dynamic peak (V) | Weak-feedback peak (V) |
|---|---|---|---|
| 1 | 0.00263 | 0.01208 | 0.01641 |
| 5 | 0.00375 | 0.00814 | 0.01172 |
| 20 | 0.06041 | 1.04148 | 0.99972 |

Under the strong 20 fF disturbance, the static output peaks at only 0.06041 V, while dynamic and weak-feedback outputs reach approximately 1.04148 and 0.99972 V, producing erroneous highs. Weak feedback does not reject arbitrarily strong disturbances. The small 1/5 fF peaks need not be monotonic because the measurement includes different injection phases and restoration dynamics.

## 5.1 Frequency boundary and reset

| Period (ns) | Frequency (GHz) | Static correct/8 | Dynamic correct/8 |
|---|---|---|---|
| 2 | 0.5000 | 8 | 8 |
| 1.8 | 0.5556 | 8 | 8 |
| 1.6 | 0.6250 | 8 | 8 |
| 1.4 | 0.7143 | 4 | 8 |
| 1.2 | 0.8333 | 4 | 8 |
| 1 | 1.0000 | 0 | 4 |

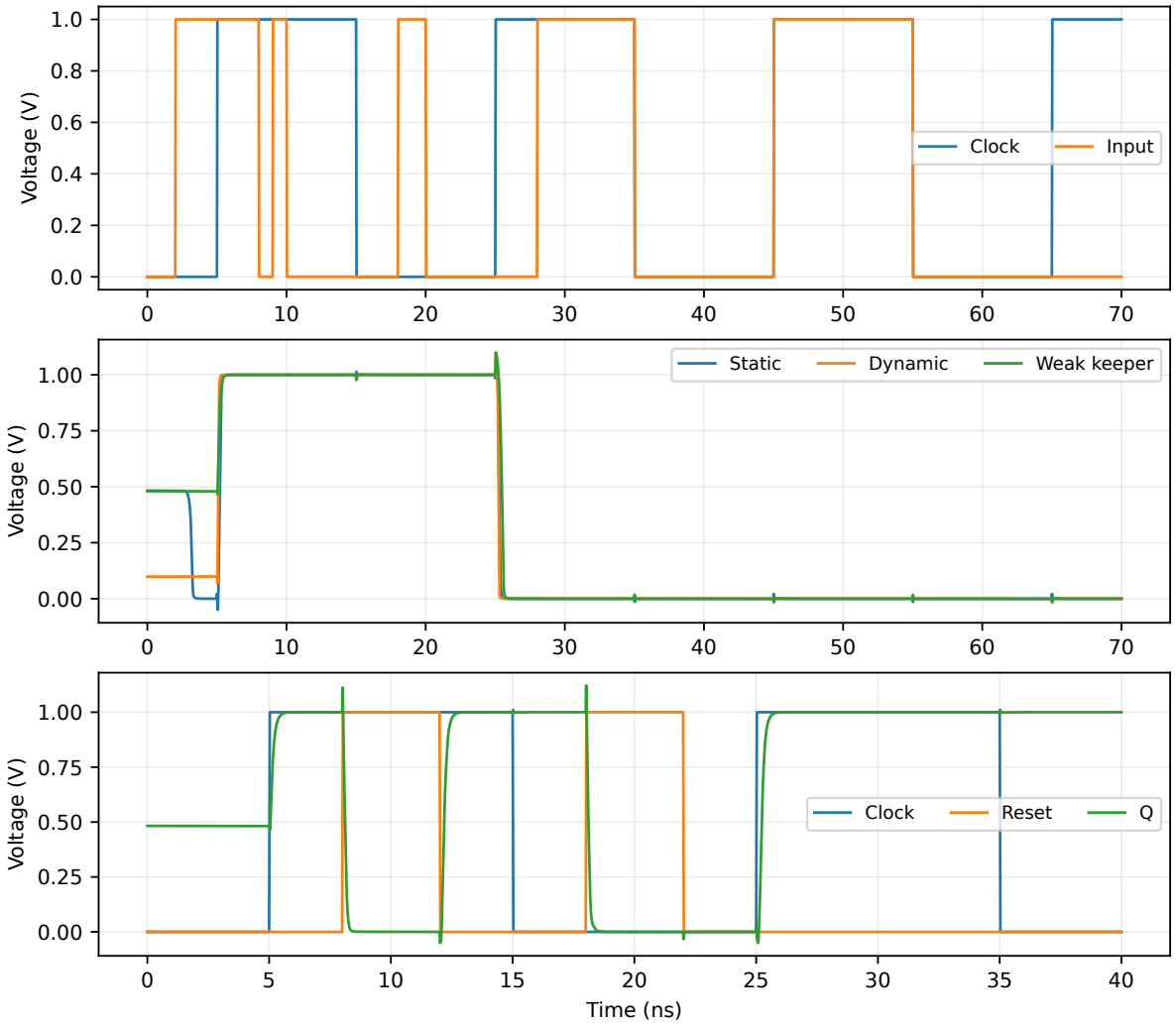



Figure 1: The upper plots show data hazards and the three DFF outputs; the lower plot shows asynchronous reset in both clock phases.

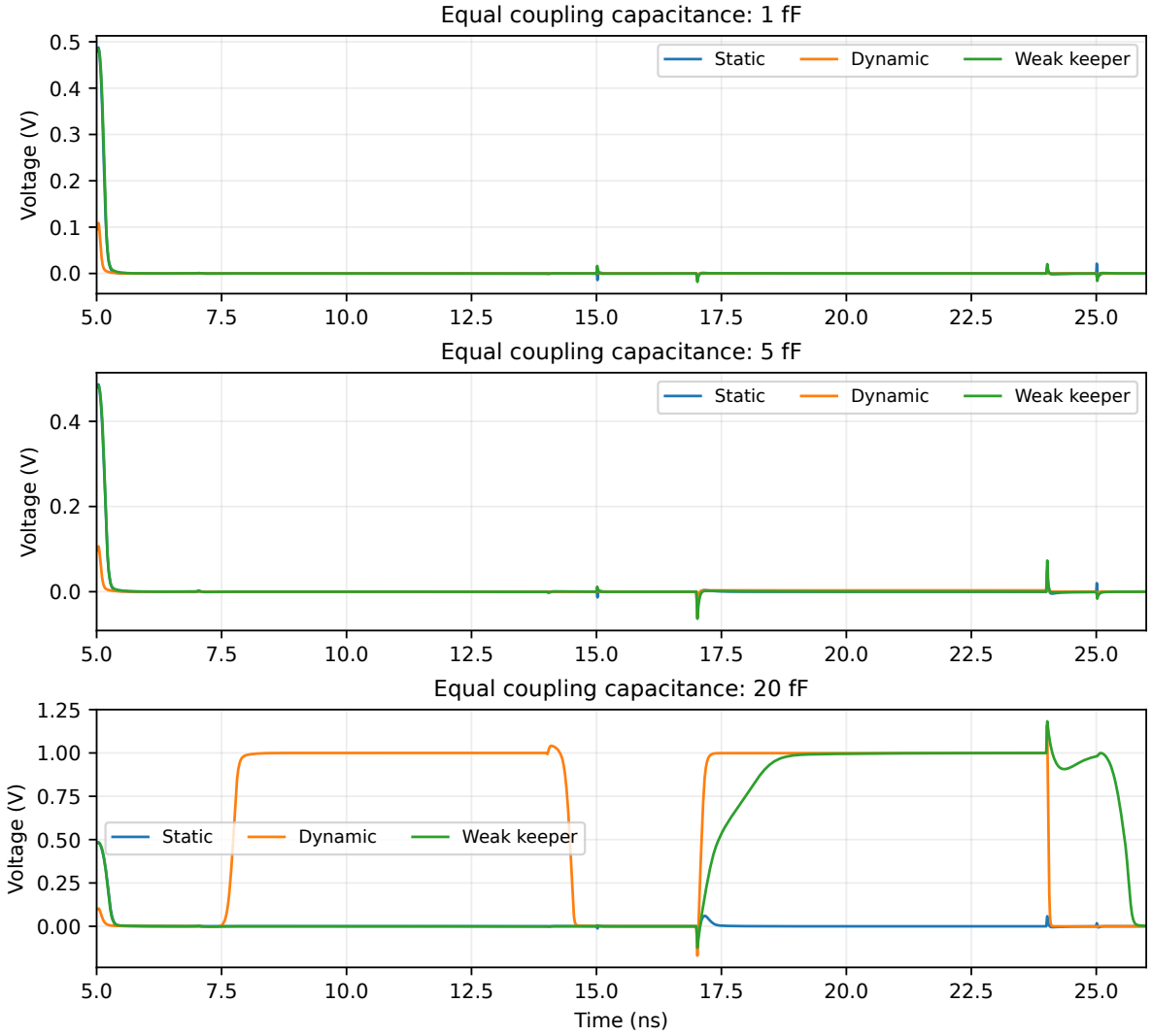



Figure 2: Identical capacitors and aggressors couple into the master/slave storage nodes with input fixed at 0.

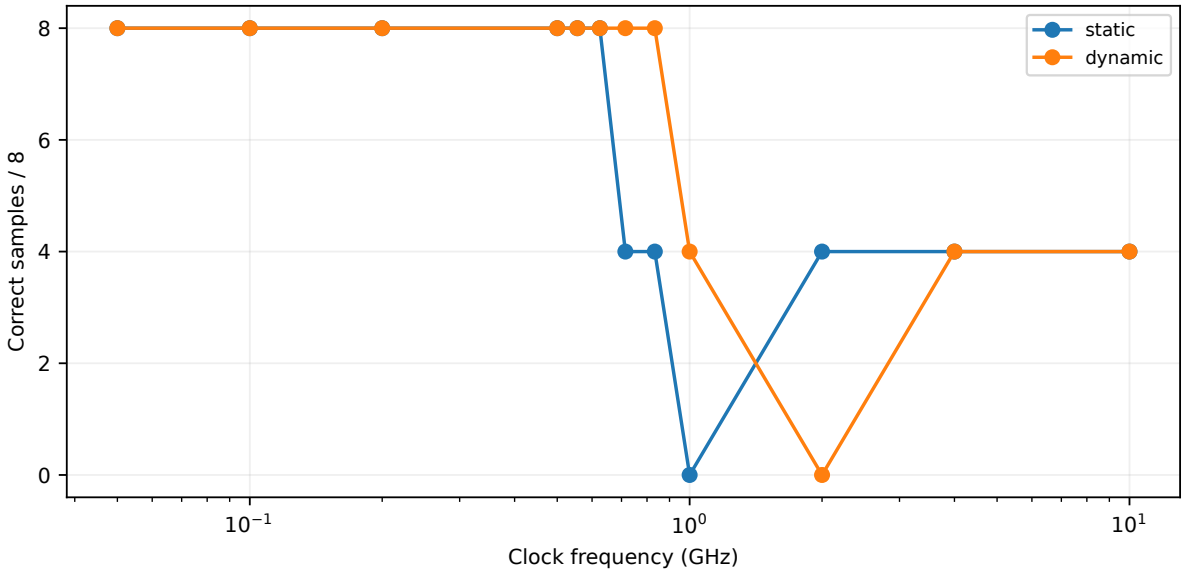



Figure 3: Fixed-time interpretation for 12 tested periods; connecting lines guide the eye and do not verify continuous frequencies.

The clock begins rising at $0.25T$ and outputs are sampled at $0.49T$, allowing about $0.24T$ to settle. The highest tested frequencies with 8/8 correct samples are 0.625 GHz (1.6 ns) for static and 0.8333 GHz (1.2 ns) for dynamic. Adjacent failing points are 0.7143 GHz (1.4 ns) and 1 GHz, respectively. These are finite-sweep boundaries under the stated load, 20 ps clock edge, and observation deadline, not unconditional $f_{\max}$ values.

Output during reset in high and low clock phases is 0.0001976 V and 3.395 $\mu$V, respectively; recapture after release gives 0.999979 V.

## 6 Discussion, differences, and unverified scope

Q1: overlapping conduction of complementary clocks can make master and slave transparent simultaneously and allow race-through. Q2: slow edges lengthen partial conduction and contention and alter available settling time, so nominal period alone is insufficient. Q3: setup allows the master storage node to settle before closing; hold prevents new data from crossing before closure completes. Both depend on clock slope, TG delay, and regeneration, rather than zero-time ideal logic. Continuous setup/hold and metastability extraction is not claimed; Lab21 separately measures slew and skew. Equal output loads and equal injected disturbances are used to avoid attributing mismatched tests to noise immunity.